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5 (a). $L_1 = \lim_{x \to k} \frac{x^2 - k^2}{x - k}$ $\boxed{\text{Dik: } k = 3,\ a = 1,\ b = 1}$

$L_1 : \lim_{x \to 3} \frac{x^2 - 9}{x - 3}$

$x^2 - 9 : (x-3)(x+3)$

$\lim_{x \to 3} : \frac{(x-3)(x+3)}{x-3}$

$: x + 3$

$L_1 = 3 + 3 = 6$ //

• $L_2 = \lim_{x \to 0} \frac{\sqrt{x+16} - 4}{x}$

$\frac{\sqrt{x+16} - 4}{x} \cdot \frac{\sqrt{x+16} + 4}{\sqrt{x+16} + 4}$

$= \frac{(x+16) - 16}{x(\sqrt{x+16} + 4)}$

$= \frac{x}{x(\sqrt{x+16} + 4)} = \frac{1}{\sqrt{x+16} + 4}$

$x = 0 \rightarrow L_2 = \frac{1}{4+4} = \frac{1}{8}$ //

• $L_3 = \lim_{x \to \infty} \frac{(a+2)x^3 + bx + 7}{x^3 - x + k}$

$\frac{(1+2)x^3 + x + 7}{x^3 - x + 3}$

$= \frac{3x^3 + x + 7}{x^3 - x + 3} : x^3$

$= \frac{3 + \frac{1}{x^2} + \frac{7}{x^3}}{1 - \frac{1}{x^2} + \frac{3}{x^3}}$

Saat $\frac{1}{x^2} \to 0,\ \frac{1}{x^3} \to 0$ $= \frac{3 + 0 + 0}{1 - 0 + 0}$

$L_3 = 3$ //

(c) $L_4 = \lim_{x \to 0} \frac{\sin((a+1)x) \cdot \tan((b+2)x)}{x^2}$

$L_4 : \lim_{x \to 0} \frac{\sin(2x) \cdot \tan(3x)}{x^2}$

$\tan(3x) = \frac{\sin(3x)}{\cos(3x)}$

$\lim_{x \to 0} \frac{\sin(2x)\sin(3x)}{x^2 \cos(3x)}$

$= \lim_{x \to 0} \left(\frac{\sin(2x)}{2x} \cdot \frac{\sin(3x)}{3x} \cdot \frac{6x^2}{x^2 \cos(3x)} \right)$

$= \lim_{x \to 0} \left(\frac{\sin(2x)}{2x} \right) \left(\frac{\sin(3x)}{3x} \right) \left(\frac{6}{\cos(3x)} \right)$

$\lim_{u \to 0} \frac{\sin u}{u} = 1$

$\lim_{x \to 0} \frac{\sin(2x)}{2x} = 1$

$\lim_{x \to 0} \frac{\sin(3x)}{3x} = 1$

$L_4 = 1 \cdot 1 \cdot \frac{6}{1} = 6$ //

(d). $R(n) = \frac{100n + k}{n + (a+b+1)}$

$R(n) = \frac{100n + 3}{n + 3}$

$\lim_{n \to \infty} \frac{100n + 3}{n + 3} : n$

$= \frac{100 + \frac{3}{n}}{1 + \frac{3}{n}}$ $\boxed{\frac{3}{n} \to 0}$

$\lim_{n \to \infty} R(n) = \frac{100 + 0}{1 + 0} = 100$ //

(3) $\lim_{n \to \infty} \left(\sqrt{n^2 + kn} - \sqrt{n^2 + bn} \right)$

$\lim_{n \to \infty} \left(\sqrt{n^2 + 3n} - \sqrt{n^2 + n} \right)$

$\left(\sqrt{n^2 + 3n} - \sqrt{n^2 + n} \right) \frac{\sqrt{n^2 + 3n} + \sqrt{n^2 + n}}{\sqrt{n^2 + 3n} + \sqrt{n^2 + n}}$

$\frac{(n^2 + 3n) - (n^2 + n)}{\sqrt{n^2 + 3n} + \sqrt{n^2 + n}} = \frac{2n}{\sqrt{n^2 + 3n} + \sqrt{n^2 + n}}$

$= \frac{2n}{\sqrt{n^2(1 + \frac{3}{n})} + \sqrt{n^2(1 + \frac{1}{n})}}$

$= \frac{2n}{n\sqrt{1 + \frac{3}{n}} + n\sqrt{1 + \frac{1}{n}}} = \frac{2}{\sqrt{1 + \frac{3}{n}} + \sqrt{1 + \frac{1}{n}}}$

$\frac{3}{n} = 0 \rightarrow \frac{1}{n} = 0 \rightarrow \frac{2}{\sqrt{1} + \sqrt{1}}$

$= \frac{2}{2}$

$= 1$ //

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Interpretasi Bisnis

Dua cara perhitungan nilai pelanggan tersebut pada akhirnya memberikan hasil yang hampir sama. Walau di awal terlihat berbeda, semakin lama selisihnya makin kecil dan mendekati 1. Ini menunjukkan bahwa dalam jangka panjang kedua model sama² bisa digunakan karena hasil prediksinya tidak jauh berbeda.

(F)

$$C(x) = \begin{cases} (x^2-k^2)/(x-k), & x < k \\ p, & x = k \\ ax+q, & x > k \end{cases}$$

Tentukan nilai p dan q

- $x < k$

$$\frac{x^2-k^2}{x-k}$$

$$x^2-k^2 = (x-k)(x+k)$$

$$\frac{(x-k)(x+k)}{x-k} = x+k$$

$$\lim_{x \to k} C(x) = k+k = 2k$$

- $x = k$

$C(k) = p$

$p = 2k$ //

- $x > k$

$C(x) = ax+q$

$\lim_{x \to k^+} C(x) = ak+q$

$ak + q = 2k$

$q = 2k - ak$

$q = k(2-a)$ //

$p = 2k = 2 \cdot 3 = 6$

$q = k(2-a) = 3(2-1) = 3$ //

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